

BATHMARAJ N

Senior AI / ML Engineer

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 10 + years of experience and 5+ years in AI/ML and Generative AI. Specialized in designing and deploying production-grade GenAI platforms using Azure OpenAI, RAG, and Agentic AI. Proven expertise in building scalable, cost-optimized, and observable AI systems serving enterprise users. Strong leadership experience in solution architecture, technical decision-making, and mentoring engineers. Demonstrated impact through automation, cost reduction, and delivery acceleration.

TECHNICAL SKILLS

Generative AI: Azure OpenAI (GPT-4, GPT-5), RAG, Agentic AI, Prompt Engineering, LLM Evaluation

Cloud & DevOps: Azure AI Foundry, Azure AI Search, Azure Functions, App Services, Azure Monitor, Azure DevOps

Frameworks & Tools: LangChain, LangGraph, LangSmith, Flask, Docker, Langfuse

Data & Storage: PostgreSQL (pgVector), FAISS, Blob Storage, SharePoint, Confluence, Neo4J

Programming: Python, SQL

ML & NLP: Computer Vision, Document AI, OCR, Embeddings

Observability: Grafana, New Relic, Prometheus, Azure Monitor, Guardrails

Optimization: Cost optimization, latency tuning, token usage optimization

Version Control: Git, GitHub

PROFESSIONAL EXPERIENCE

Senior Digital Engineer (Associate Architect) – Sonata Software | July 2025
– Present

CAMP AU – GenAI Child Care Auto Mail Drafting System

- Designed and deployed a GenAI-powered automated email drafting system serving ~3,00+ childcare staff users.
- Automated manual email drafting workflows, reducing average response preparation time by ~60%.
- Built multi-query AI response generation using Azure GPT-5 with contextual grounding.
- Implemented enterprise-scale RAG

over SharePoint, Confluence, Blob Storage, and PostgreSQL (pgVector), indexing 200+ enterprise files (~8,000 text chunks) across 6 business data sources. - Processed ~1,500 automated emails per week (~200–250 per day) with 99.5% uptime. - Implemented LLM evaluation pipelines to monitor hallucination rate and response relevance. - Reduced Azure OpenAI token cost by ~28% through prompt optimization and caching. - Improved average response latency from ~6.2s to ~3.9s using async orchestration and retrieval tuning. - Led architecture decisions for governance, observability, and CI/CD using Azure AI Foundry and Azure DevOps.

CSOD – Implementation Blueprint & Project Intelligence Platform

- Delivered AI-driven project intelligence features used by ~1,500+ enterprise users. - Integrated Azure AI Search for intelligent document discovery over 200+ SharePoint and enterprise documents. - Executed Alembic migrations and schema changes with zero-downtime releases. - Deployed Azure Functions and App Services with Bitbucket-based CI/CD. - Reduced manual reporting effort by ~40% through automation.

Senior AI/ML Engineer (Associate Architect) – Genworx.AI | Feb 2025 – June 2025

- Architected as an Associate enterprise GenAI platforms using Azure OpenAI (GPT-4 & GPT-5).
- Designed Agentic AI workflows with LangGraph and LangSmith for traceability and guardrails.
- Built scalable RAG pipelines over structured and unstructured data (~280+ enterprise documents).
- Served ~300+ AI requests/day across internal tools.
- Implemented LLM evaluation dashboards to track response quality and drift.
- Reduced token usage cost by ~22% via prompt compaction and retrieval filtering.
- Tuned retrieval + generation latency by ~35%.
- Led technical design reviews and mentored 3 junior engineers.

Machine Learning Engineer (Lead Engineer) – Lumina Datamatics | Mar 2022 – Feb 2025

- Mentored 4 engineers and conducted technical design reviews.
- Led feasibility analysis and end-to-end solution design for AI/ML and GenAI initiatives.
- Developed document intelligence pipelines using OCR, YOLO-based vision models, and NLP.
- Processed ~2,400 pages/day (~72,000 pages/month) across publishing workflows.
- Deployed Dockerized Flask services handling ~100+ requests/day.
- Reduced manual data extraction effort by ~50%.

- Collaborated with business stakeholders to define architecture and delivery roadmap.

Python Developer – Auditics Pvt Ltd | Feb 2021 – Feb 2022

- Automated financial audits and document workflows using Python and OCR.
- Built OCR-based invoice and document processing pipelines.
- Reduced manpower dependency by automating weekly audit workloads.
- A task that required **4 people working for 1 full week** was reduced to **~45 minutes/day** through OCR automation.
- Delivered ~85–90% reduction in manual effort and turnaround time.
- Built NLP-based document classification models.

EARLY CAREER EXPERIENCE

Testing Engineer – Prostarm Info Systems Pvt Ltd | May 2021 – Mar 2022

- Performed testing and validation of PCBs, PSDR units, and power electronics, ensuring quality compliance.
- Identified defects, conducted root cause analysis, and collaborated with R&D to improve product reliability.

PCB Engineer – Numeric India (Legrand Group) | May 2015 – Mar 2021

- Quality assurance and process optimization for UPS and power electronics manufacturing.
- Improved defect detection efficiency by ~20% through process standardization.

EDUCATION

B.E. Electronics & Communication Engineering – Alpha College of Engineering (2011–2014)

Diploma in Electronics & Communication Engineering – Cheyyar Polytechnic College (2009–2011)

CERTIFICATIONS

Artificial Intelligence & Machine Learning – Soft Logic Systems

Python Certification – Soft Logic Systems